MALNAD COLLEGE OF ENGINEERING HASSAN
Department of Computer Science and Engineering



Main-Project Phase II-23CS703

"Hybrid Quantum-Classical Approach"

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Agenda :

- Introduction
- Problem Statement
- System design and development
- Identification of appropriate tools
- Conclusion
- References

Introduction to Heart Disease

Early Prediction and Its Importance

Heart Disease

Heart disease is a leading cause of death worldwide, making **early detection** essential for effective treatment and management, potentially saving countless lives through timely interventions.

Machine Learning

Employing machine learning techniques allows for the **analysis of complex** medical data, facilitating pattern recognition and improving the accuracy of early heart disease predictions significantly.

Quantum Computing

Quantum computing introduces innovative methods that enhance computational capabilities, providing new opportunities for **solving intricate** problems in medical data analysis and improving prediction models.



Problem Statement

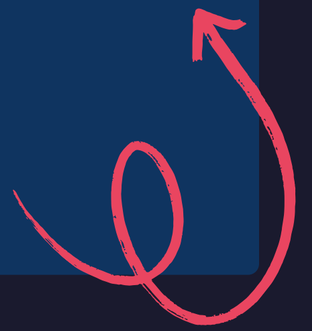
Addressing Early Heart Disease Prediction

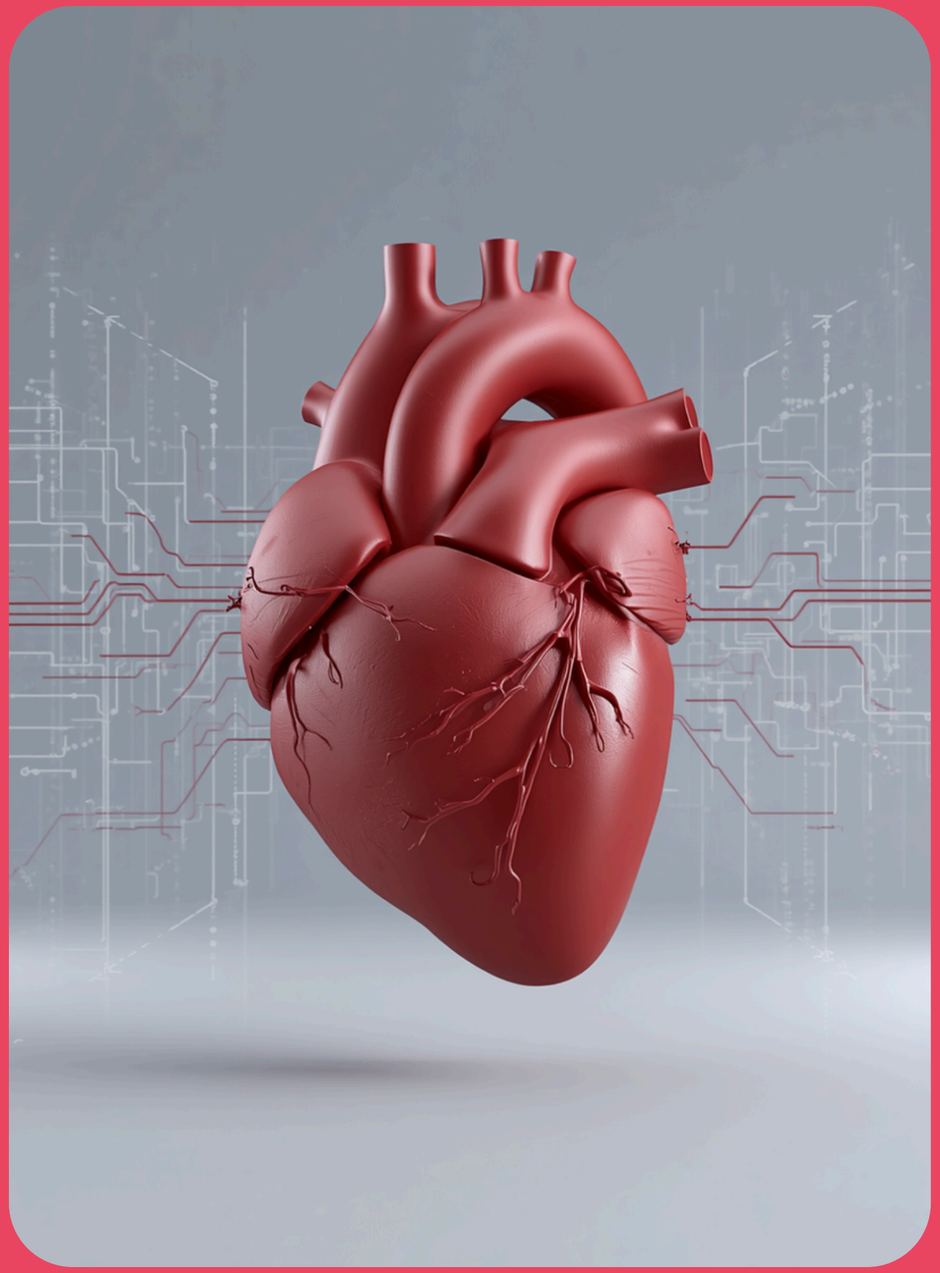
Traditional Diagnosis

Traditional diagnostic methods rely heavily on **medical expertise** and multiple clinical measurements, leading to potential delays in identifying patients at risk for heart disease.

Computational Methods

There is a growing need for **computational approaches** that can efficiently analyze patient data, aiding in early risk prediction while reducing reliance on subjective clinical judgment.





System Design and Development

Dataset Overview

Heart Disease Prediction Dataset Details

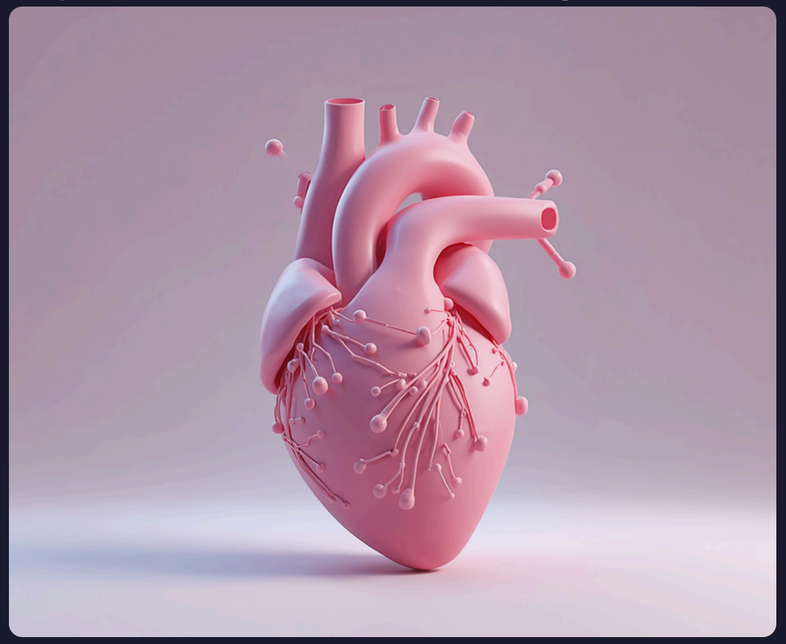
The dataset contains **270 patient records** with 13 input features, enabling analysis and preprocessing to enhance the prediction of heart disease risk through machine learning techniques.

System Design and Development

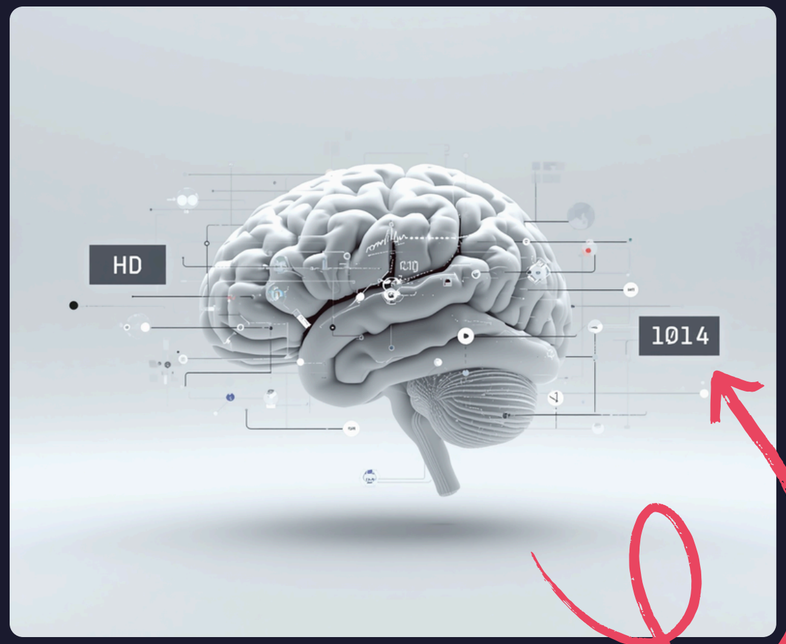
Classical Machine Learning Models

Performance Evaluation Results Overview

We implemented five classical machine learning models to predict heart disease, achieving varying levels of accuracy. The best model, SVM, reached an impressive **83.31% accuracy**.



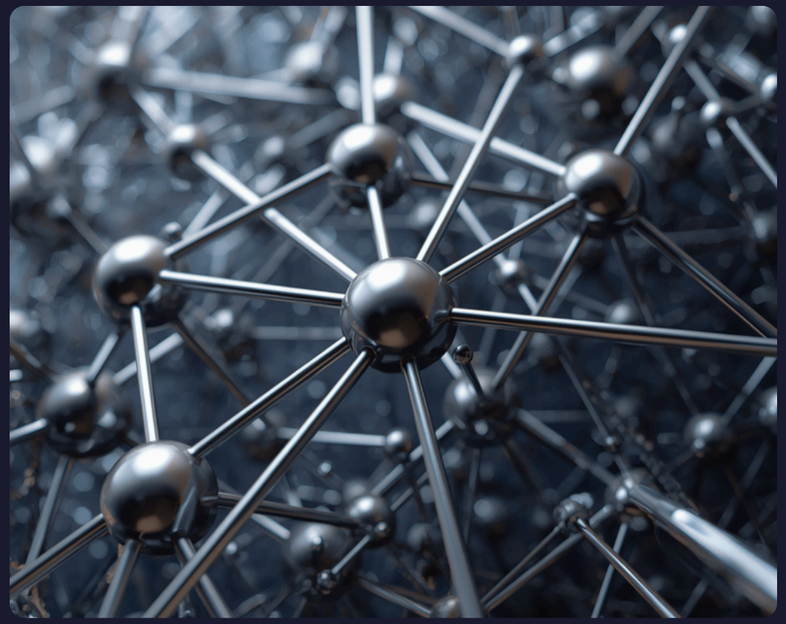


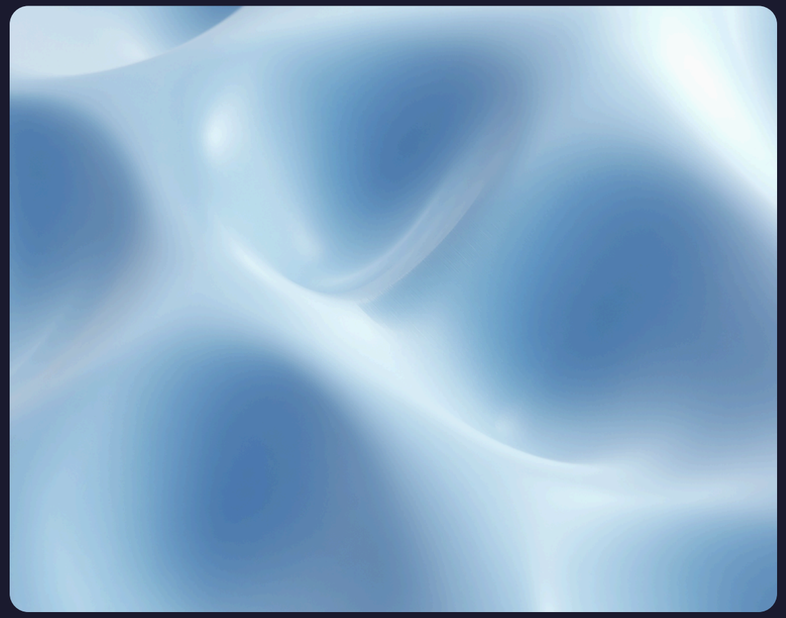


Overview of Quantum Machine Learning

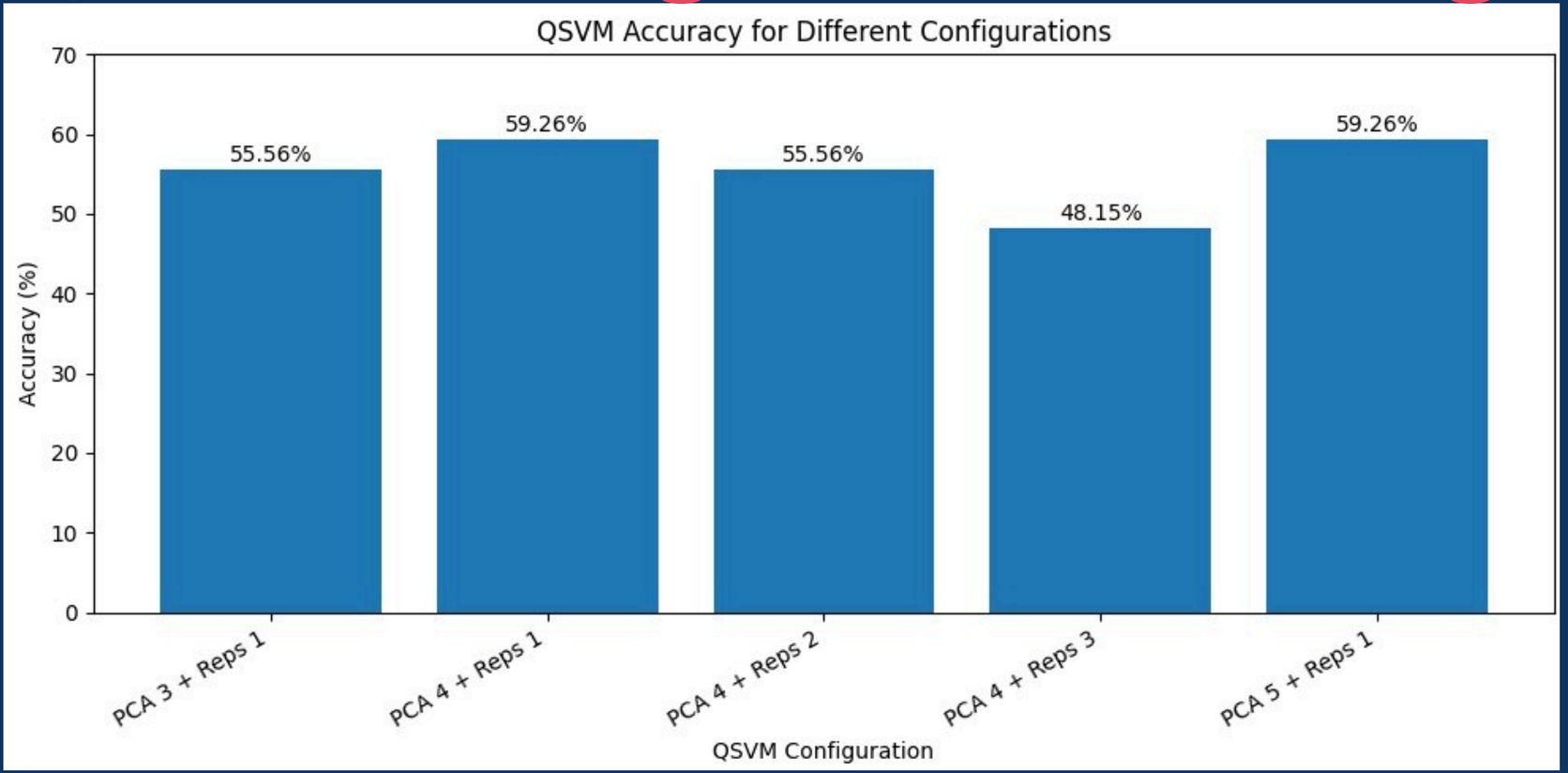
Harnessing Quantum Principles for Prediction

Quantum machine learning leverages quantum computing to process **complex data** more efficiently, utilizing quantum states and kernels to enhance predictive capabilities for heart disease analysis.





QSVM Configuration Findings



Multiple configurations were tested; the best performance achieved was with PCA 4 features and 1 repetition, resulting in a notable accuracy of **59.26%** for the quantum model.

QSVM Cross-Validation Outcomes

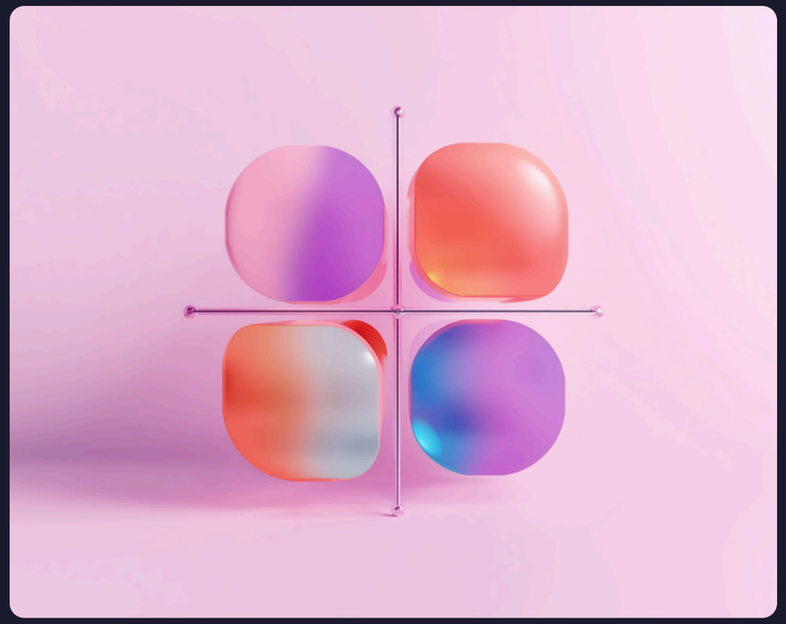
Evaluating Performance Across Multiple Folds

The QSVM model underwent 5-fold cross-validation, yielding a mean accuracy of **54.81%**, highlighting the model's potential yet indicating room for performance improvement compared to classical approaches.

QSVM Test Results Overview

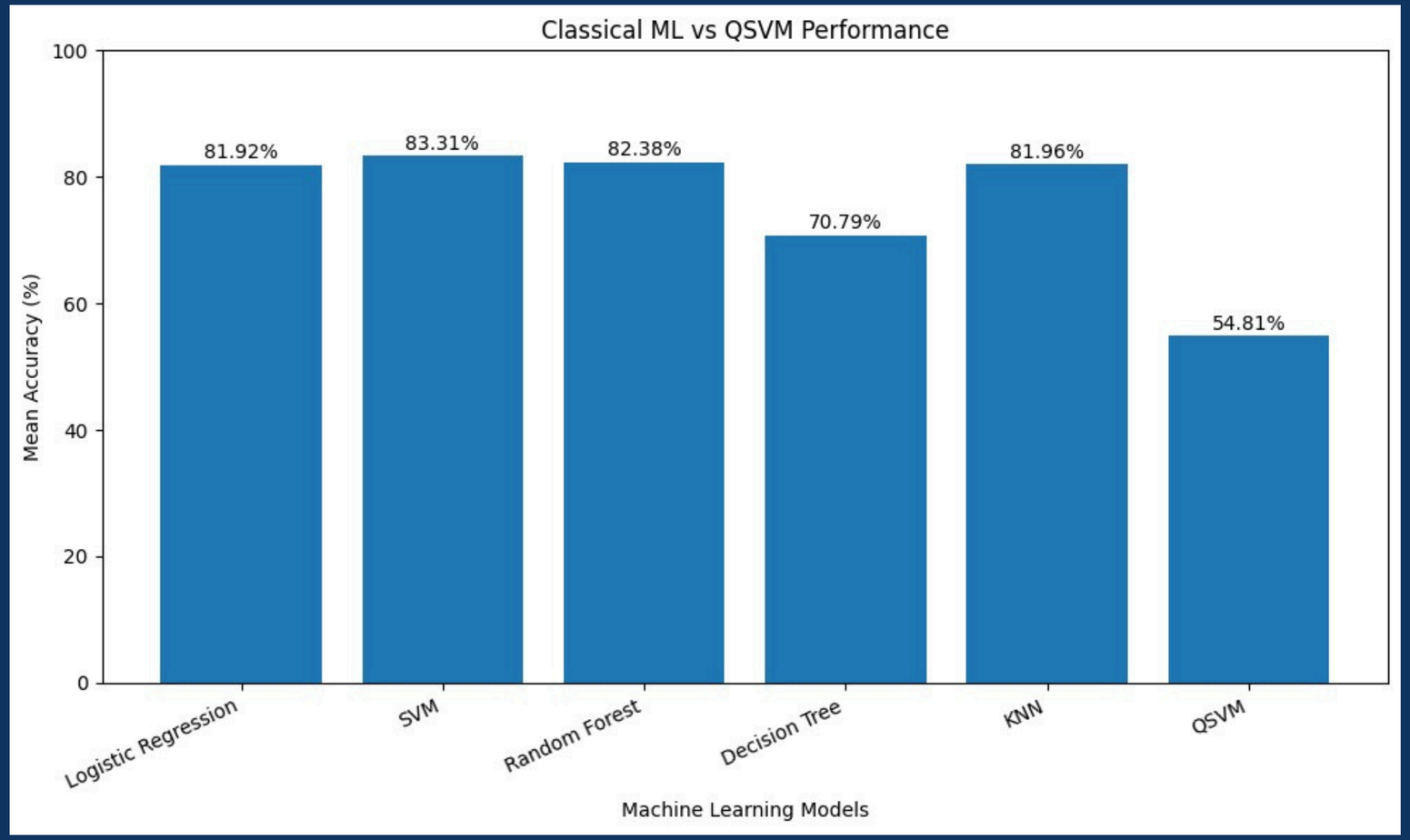
Performance Assessment and Analysis

The QSVM achieved a test accuracy of **59.26%**, utilizing a confusion matrix to analyze performance metrics, highlighting its potential despite not surpassing classical models.





Performance Comparison



Performance Comparison

Analyzing Classical and Quantum Approaches

Classical Performance

The best classical model, **SVM**, achieved an impressive accuracy of **83.31%**, demonstrating its effectiveness in predicting heart disease based on the provided dataset.

Quantum Performance

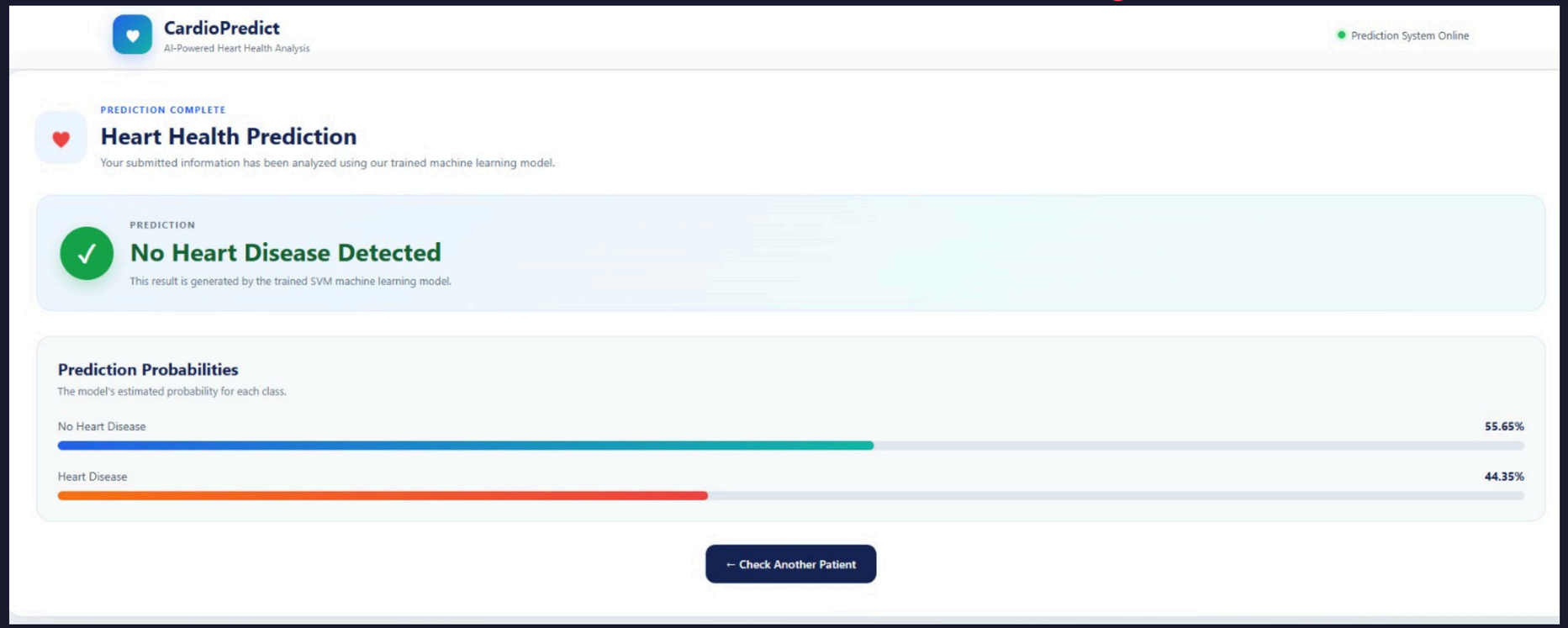
The QSVM achieved a mean accuracy of **54.81%**, showing that while quantum techniques can be applied, they currently lag behind classical methods on this dataset.

Key Insights

Despite the lower performance of QSVM, the project illustrates the potential for future quantum machine learning advancements and encourages ongoing exploration of hybrid approaches.



Web-Based Prediction System



Our developed application offers a user-friendly interface for heart disease predictions, using trained models to provide real-time results based on patient inputs without storing sensitive data.

Identification of Appropriate Tools

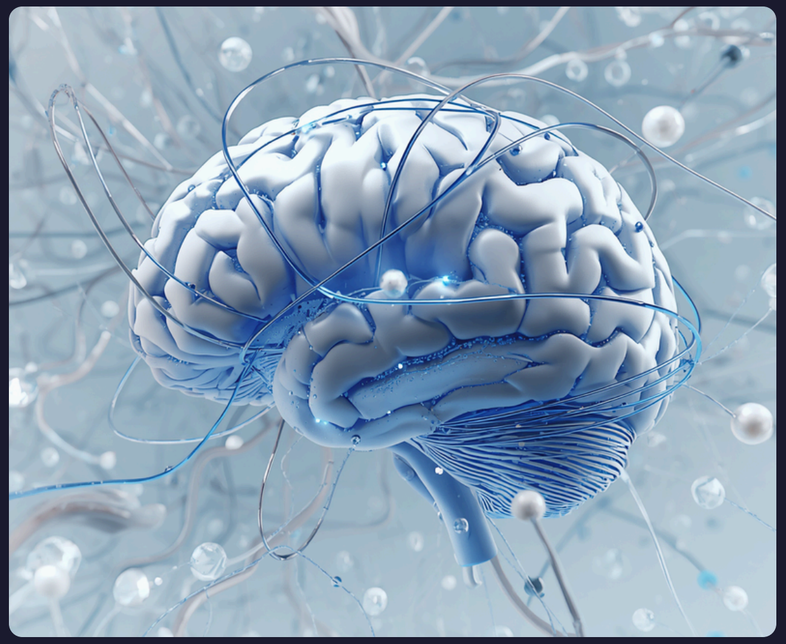
Component	Tool/Technology
Programming	**Python**
ML Development	**Scikit-learn**
Quantum ML	**Qiskit / Qiskit Machine Learning**
Data Processing	**Pandas, NumPy**
Dimensionality Reduction	**PCA**
ML Model	**SVM**
Quantum Model	**QSVM**
Web Application	**Web-based frontend + backend**
Development	**Jupyter Notebook / VS Code**

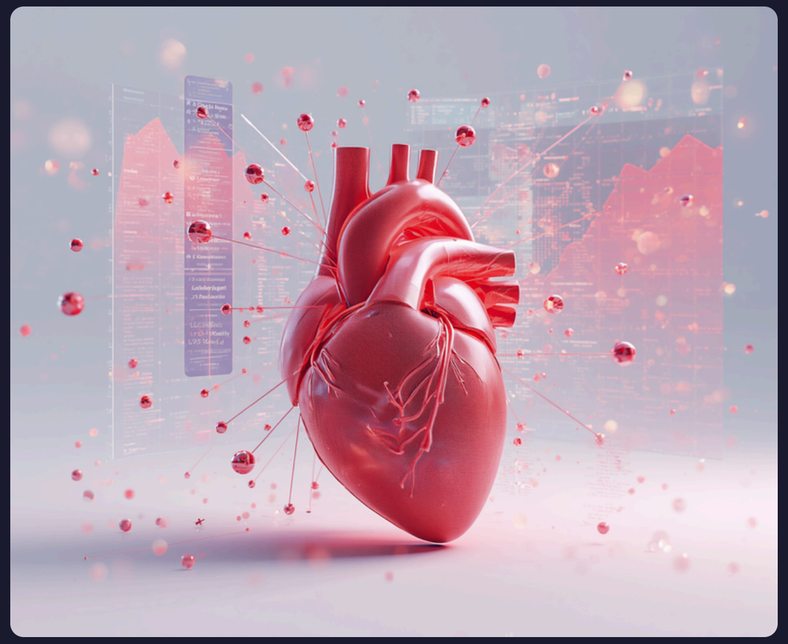


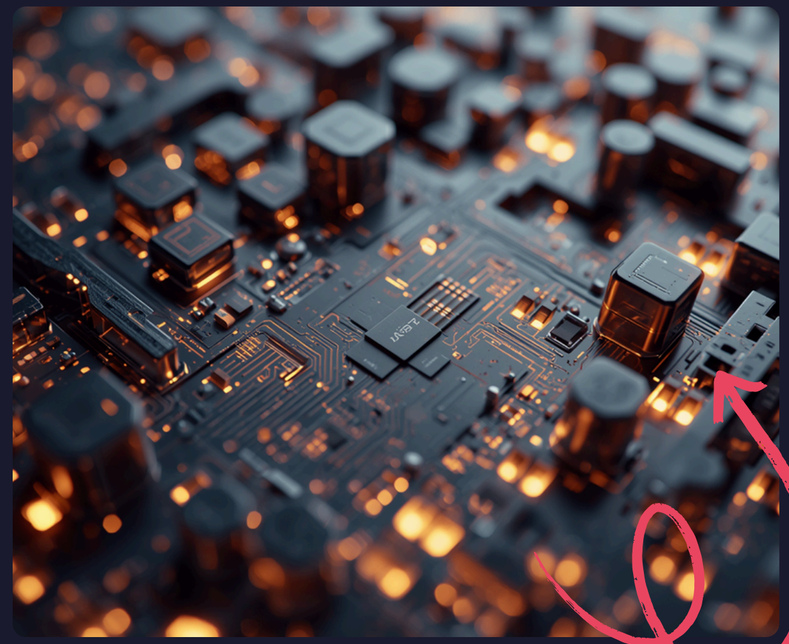
Final Results and Conclusion

Key Findings from the Project

The project highlights that classical machine learning, particularly SVM, outperformed quantum approaches for heart disease prediction, demonstrating the effectiveness of both methods in analyzing patient data.







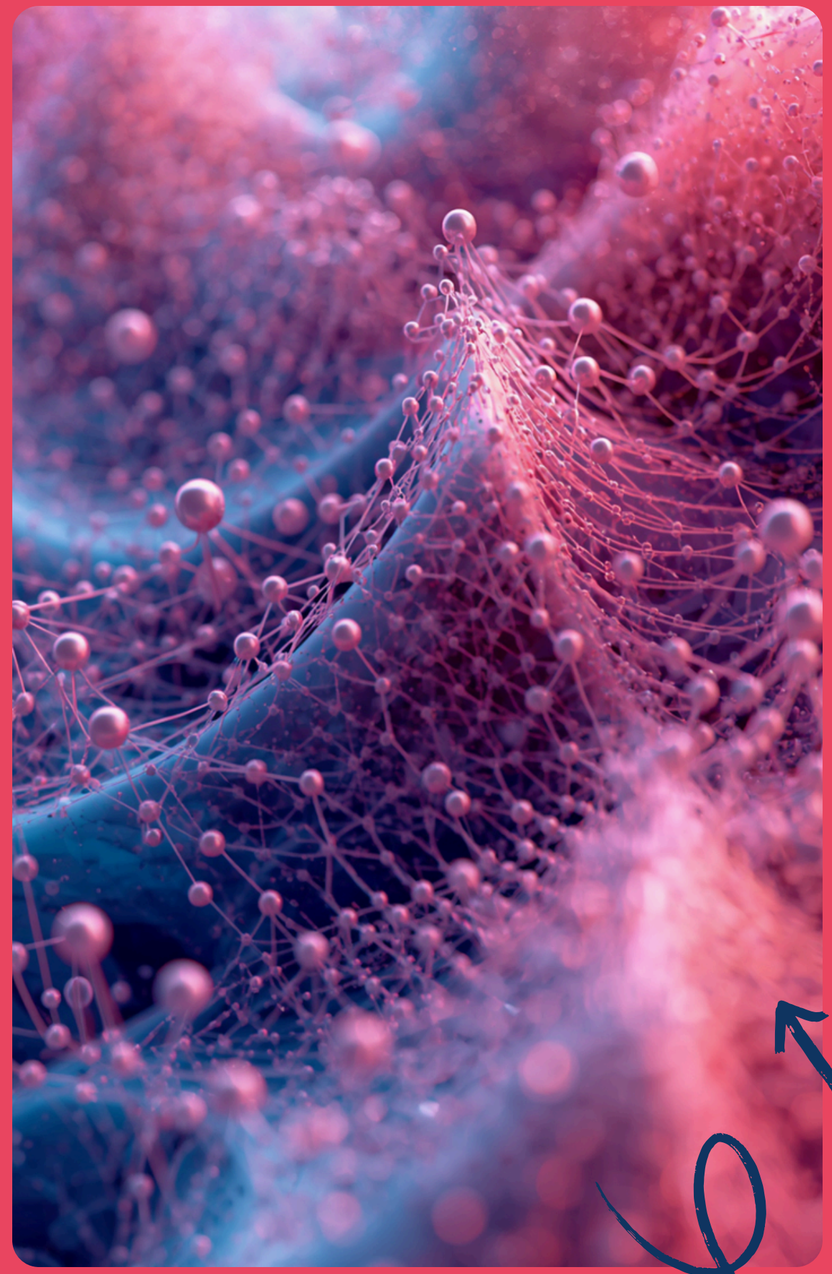
Future Research Directions

Expanding the Horizons of Heart Disease Prediction

We aim to explore larger datasets, advanced quantum techniques, and improve our web application’s features, ultimately enhancing prediction accuracy and user experience for heart disease assessment.

References

- Research papers on Machine Learning for Heart Disease Prediction.
- Research papers on Quantum Machine Learning and QSVM.
- Scikit-learn documentation — SVM and PCA.
- Qiskit documentation — quantum circuits and quantum machine learning.
- Any papers used to justify the quantum feature map and hybrid approach.



Thank You